Design Documentation:
– All parts are listed, along with their specs, McMaster code (if relevant), and fabrication details (if relevant)
– Clear illustration of the design intent of the prototype, especially the functional and moving parts. This can be sketches, CAD blow-up, or other; consider what you are trying to communicate. Annotate the illustrations appropriately for components and function/motion.
– Clear instructions of how to assemble the prototype. Think back to your reverse-engineering activity. Label components and details relevant to the assembly, and use (visually different) arrows to indicate both assembly steps and mechanism functionality.

1. [Bluetooth Speaker]:
    a.

Additional details	
Mounting Type	Tabletop Mount
Enclosure Material	Plastic
Speaker Type	Portable Speaker
Additional Features	Stereo Pairing
Color	Black
Water Resistance Level	Waterproof
Power	
Speaker Maximum Output Power	5 Watts
Sound Quality	
Frequency Response	20 KHz
Measurements	
Unit Count	1.0 Count
Item Dimensions D x W x H	2.2"D x 1.2"W x 1.4"H
Compatibility	
Compatible Devices	Smartphone

Item details	
Brand	HZDYDK
Model Number	i12
Model Name	Bluetooth Speaker
Built-In Media	Quick Start Guide
Number of Items	1
Manufacturer	HZDYDK
Warranty Description	1
Best Sellers Rank	#249 in Portable Bluetooth Speakers #2,202 in MP3 & MP4 Player Accessories
ASIN	B0G2RKPV9N
Customer Reviews	4.6 (174) 4.6 out of 5 stars
Connectivity	
Connectivity Technology	Bluetooth
Subwoofer Connectivity Technology	Wireless

2. [Zip Ties]
    a.

Black Zip Ties 8 inch 100 Pack

Skalon Co.,Ltd is a professional manufacturer of nylon cable ties.
We will continue to improve the product quality, provide top grade products and the best service for our customers.
Make the best use of your purchase!
With 100 zip ties in each package, you don't have to worry about running out in the middle of your project.

High Quality
Skalon black zip ties are made of UV resistant halogen free nylon 66.
Operation temperature is -40℉to 185℉with UL94V-2 flammability.

Higher Tensile Strength
The tensile strength of our zip ties is 40 pounds, higher than others.
Our zip ties'width is 0.14 inch, wider than industry standards.
You can use our wire ties with confidence without worrying they will break.

Self-locking Design
Skalon zip ties can be easily locked in, tightly and securely.
One-piece construction for consistent performance and reliability.

Multi-purpose Applications
Skalon plastic ties can be used to tie cords and cables for electronics.
8 inch zip ties can be used to organize electrical wires and PC or laptop cords.
You can use them to repair garden fences, hang LED lights, etc.

Warranty:
Skalon zip ties have a 1-year warranty period.
Our customer will receive a 100% refund if any quality issues under warranty period.

Tips:
1. Please keep out of reach of the children.
2. Please keep it sealed for long-term use.
3. Please use zipties in proper temperatures and tensile strength.
Package Includes:
8 Inch Zip Ties Black 100 Pack.

3. Duct Tape:
    a. 3 pack

Item details	
Brand Name	Amazon Basics
Number of Items	1
Recommended Uses For Product	Packaging, Indoor, Outdoor, Arts And Craft, Painting, Plumbing, Electrical, Sublimation
Surface Recommendation	Stone, Cardboard, Plastic, Wood, Metal
Global Trade Identification Number	00672352802726
Manufacturer	Amazon
UPC	840324410990 672352802726
Item Weight	9.18 ounces
Unit Count	3.0 Count
Best Sellers Rank	#1,062 in Industrial & Scientific (See Top 100 in Industrial & Scientific) #19 in Duct Tape
ASIN	B07YDSC22B

Features & Specs	
Other Special Features of the Product	Strong
Water Resistance Level	Water Resistant
Compatible Material	Cardboard, Metal, Plastic, Stone, Wood

Measurements	
Item Dimensions L x W x Thickness	1.89"W x 1.88"Th
Size Name	1.88-inch by 30-yard
Item Thickness	4.77 Centimeters

Additional details	
Color	Silver
Material Type	Vinyl

4. Methyl Salicylate:
    a.

Features & Specs	
% Assay of Chemical(s)	99
Container Type	Bottle
Item Form	Liquid

Measurements	
Volume	4 Fluid Ounces

Feedback

Would you like to tell us about a lower price?

Item details	
Brand Name	CCS CONSOLIDATED CHEMICAL & SOLVENTS
Manufacturer	CCS CONSOLIDATED CHEMICAL & SOLVENTS
UPC	712038845330
Included Components	Methyl Salicylate in Clear Plastic Bottle
Unit Count	3.3 Fluid Ounces
Best Sellers Rank	#73,200 in Industrial & Scientific (See Top 100 in Industrial & Scientific) #297 in Commercial All-Purpose Cleaners
ASIN	B07DLZTRFB

5. UV-Resistant Sure-Grip Rope
    a. McMaster Code: **3818T23**
    b. To attach our trap to trees
6. Resealable Bags
    a. McMaster Code: **1959T63**
    b. To hold SLF that we trap

<u>Design Intent:</u>

Our product is indeed to hang against a tree trunk using a strap-and-hook attachment style. A speaker carriage at the top of our product supports our vibration source whilst a funnel directs the SLF into the collection bag. The bag provides storage volume to maximize SLF capture whilst having a narrow funnel reduces the chances of escape. A key design choice was deciding between a bag or a hard shell. Although a hard shell would have been more aesthetic and increased user friendliness, the added design complexity, cost, and inability to store compactly made us select a bag. The critical functional parts are:

1. Secure hanging of our product on the tree
2. Bag can support increasing load of SLF
3. Entry of SLF through funnel; attracted by winter green oil
4. Efficient Removal/Replacement of the bag

<u>Assembly Steps:</u>

1. Fabricate the funnel, hook and speaker carriage
2. Place the speaker into the carriage
3. Attach the funnel to the connector
4. Modify the collection bag with openings for the funnel and hook and add attractant system (winter green oil)
5. Fit the bag over the funnel and speaker housing structure and secure it
6. Attach the final assembled product to the tree using the rope
7. Hang the product and ensure the bag hangs freely and the funnel remains aligned

Design Test:

for each test, indicate which part it is testing, what it is testing for, how you
perform the test, the test results, and what your conclusion is for the next iteration.
3
Design Tests:

1. Attachment Security/Anti-Slip Test

Part Tested: Hook and Tree attachment system
What is being tested? Whether the device remains in position on the tree with minimal to zero sliding and rotating.
How the test was performed:
- Attach our prototype to a street pole outside
- Add loads in increments to simulate the accumulation of SLF's
- Record Vertical slip distance over the course of 1 min and 10 mins
- Repeat the trial 3 times

Do loads of around 0.5lbs,1lbs, 1.5lbs and 2lbs
Slip distance allowed: Example: Less than 0.5 inches in 10 mins
Also helps us determine whether our prototype will sag. This helps identify our max load.

2. Rain/Drainage Test

Part Tested: Bag + Speaker System
What is being tested? Whether water accumulates in the bag and interferes with the function of the speaker.
Values to consider:
- mL of water sprayed/used
- mL of water retained after 12 hours
- Time taken for drainage
- Any blockage or functionality issues observed

3. Speaker Test

Part Tested: Speaker, Speaker module
Testing whether the speaker is secure in the 3d printed module and properly omits a 60Hz signal for attracting lanternflies.
Values to consider:
- At what rotations the speaker falls out.
- Measured frequency of speaker
- DBs of speaker

Success Criteria:
briefly (1-2 sentences) give the context of your project, then itemize your
criteria, what it is assessing, and how you will measure it. Identify at least one criterion that
is relevant to an exhibit-day demonstration of your prototype, and describe how it will be
Incorporated.

This product aims to attract and trap a high volume of spotted lantern flies through the use of certain sound frequencies and scented oils. The design of this product is a bag that is meant to be hung outdoors where it can be most effective at attracting lantern flies. Thus the criteria of testing should be making sure the speaker producing the sound works (at the required frequency) and that the prototype can withstand the elements.

Criteria 1:

The bluetooth speaker is accurately able to relay sound within a frequency range of 50-70 Hz.

Criteria 2:

The backbone and bag should be able to withstand a minimum load of 1.5*(Factor of safety of 3) = 4.5 lbs and undergo any static failure.

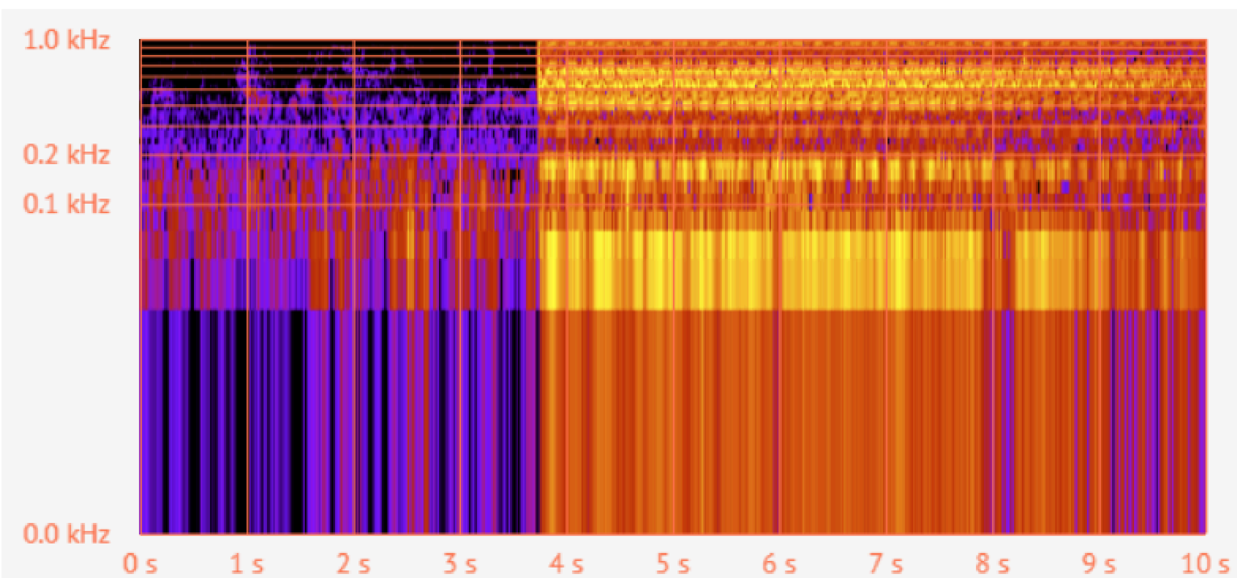
Criteria 3:

Water will be poured on the prototype in a way rain would hit it (from above) for 60 seconds and no water should be found on the speaker or in the bag.

Prototype test notes:

Criteria 1:

Fail: our speaker was able to produce a frequency of approximately 80 hertz



Criteria 2:

Fail: Our speaker carriage fractured under the load at the narrow connection point. It failed under a load of 1 additional pound of weight.

Criteria 3:

Success: The speaker carriage rain shield prevented water from entering the bag or speaker for 60 seconds

Conclusion/ Design Changes for next prototype:

Assembly of our prototype from the base components was difficult. It took too long and the process must be streamlined and some components of our product must be altered for better integration. We tested the frequency of the speaker and determined it wasn't the correct frequency.

For the speaker carriage:

There is a small pole currency holding up the hood of the speaker carriage to prevent water drainage into the speaker. After manufacturing and testing however, we have determined that this is not a mechanism that aids in preventing water drippage however it adds significant manufacturing complexity. Furthermore, the flat backplate is too long to feasibly manufacture in one part so we will be breaking it down into a flat plate with a few holes. The speaker carriage and funnel will then be modified to include hooks to latch onto the holes.

For flat support beam connecting funnel and speaker carriage:

For this prototype we used cardboard but for our next prototype we will use plastic to prevent water deterioration.

For the bag:

The bag is difficult to attach to the backbone in our prototype. For the future, we can either design or purchase clips to attach to our backbone to make it easier to clip on the bag to our design.

For the funnel:

Include a flat or recessed ridge for the zip tie to tighten onto. The funnel shape can be altered to see if it would be better at amplifying the speaker noise.

For the speaker:

The bluetooth speaker used in the test was observed to only reach a lowest frequency of 80 Hz which is about 20 Hz above our target frequency. We have supplies for our own speaker now and will be testing that. The bluetooth speaker used in our first prototype is also dependent on external devices to produce sound as well as uses a lot of power (part of the power is used for bluetooth capabilities). Using different speaker components as well as electrical circuits+components, we should be able to design a speaker system that produces our desired frequency of 60 Hz independently and is more power efficient.